

Thevenin's Theorem

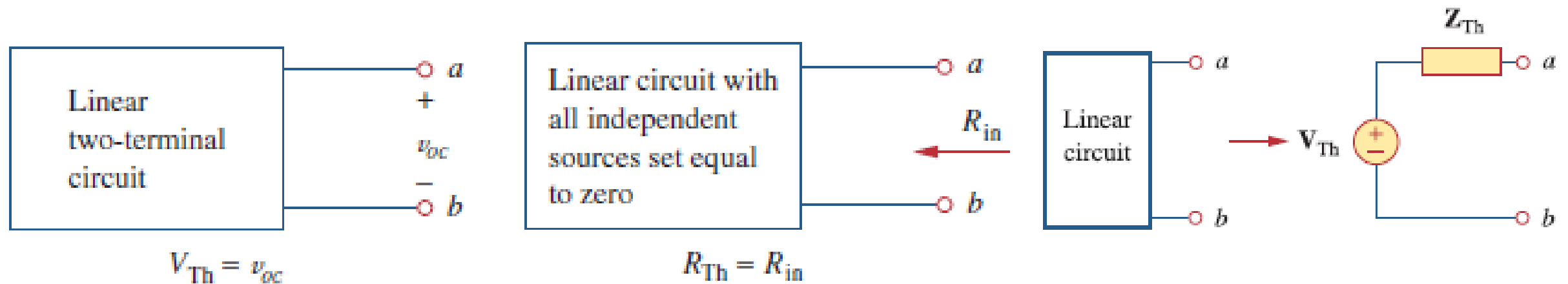
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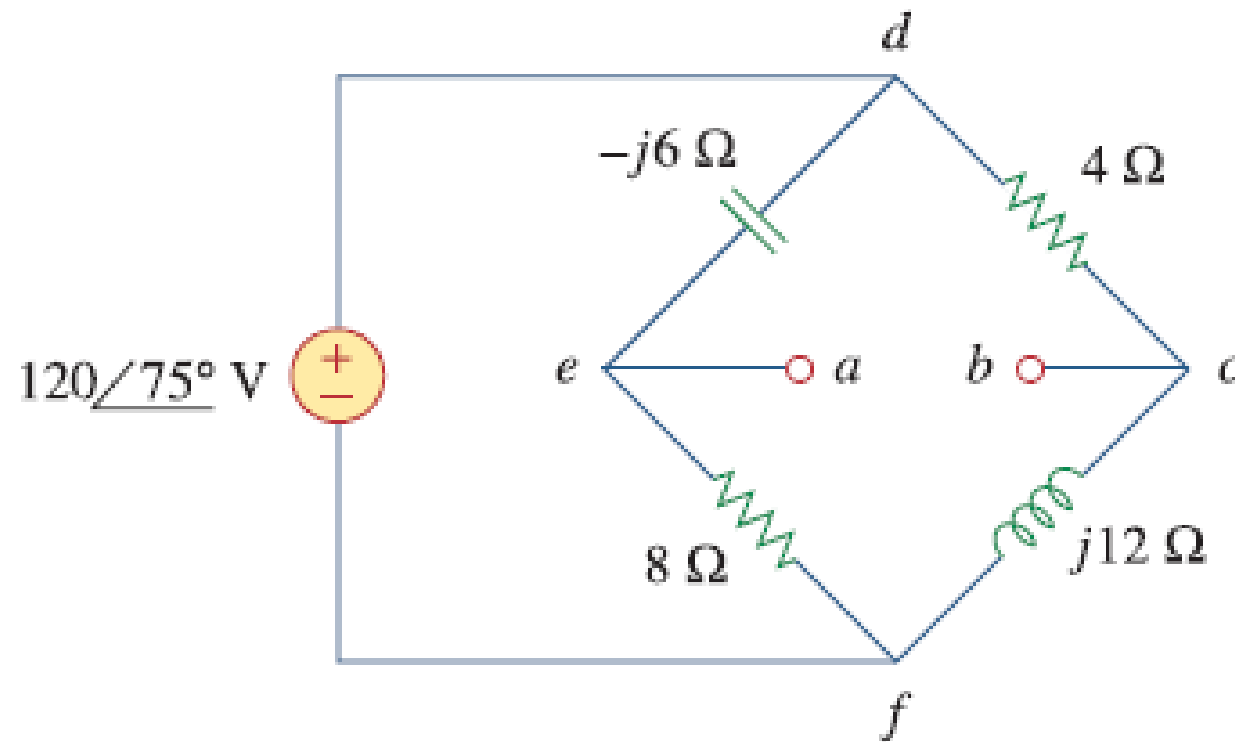
Thevenin's Equivalent Circuit

Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.



Numerical -1

Obtain the Thevenin equivalent at terminals a - b of the circuit



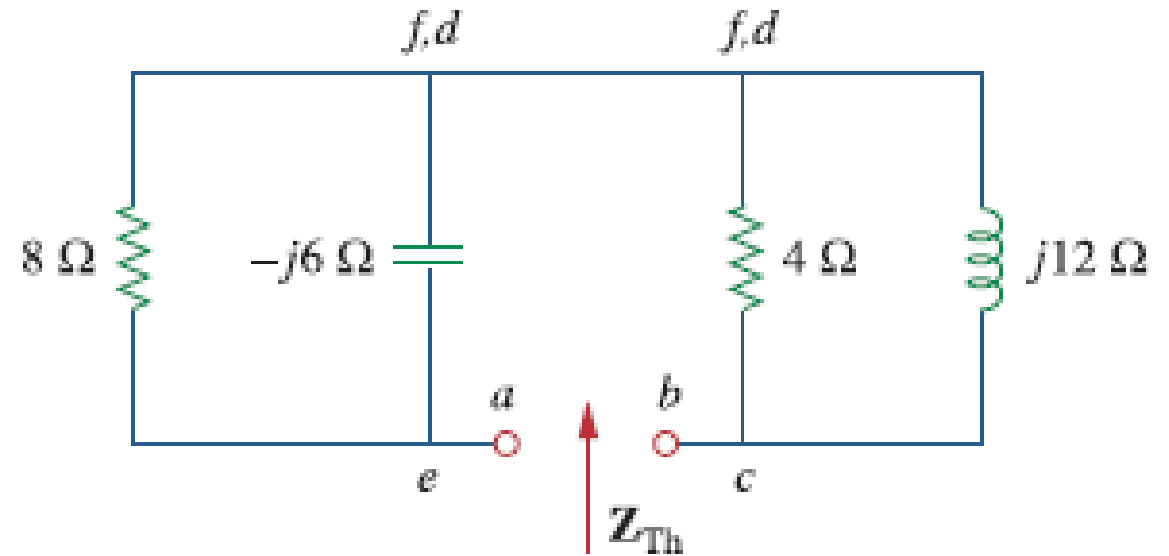
Solution:

To find R_{Th} :

$$\mathbf{Z}_1 = -j6 \parallel 8 = \frac{-j6 \times 8}{8 - j6} = 2.88 - j3.84 \, \Omega$$

$$\mathbf{Z}_2 = 4 \parallel j12 = \frac{j12 \times 4}{4 + j12} = 3.6 + j1.2 \, \Omega$$

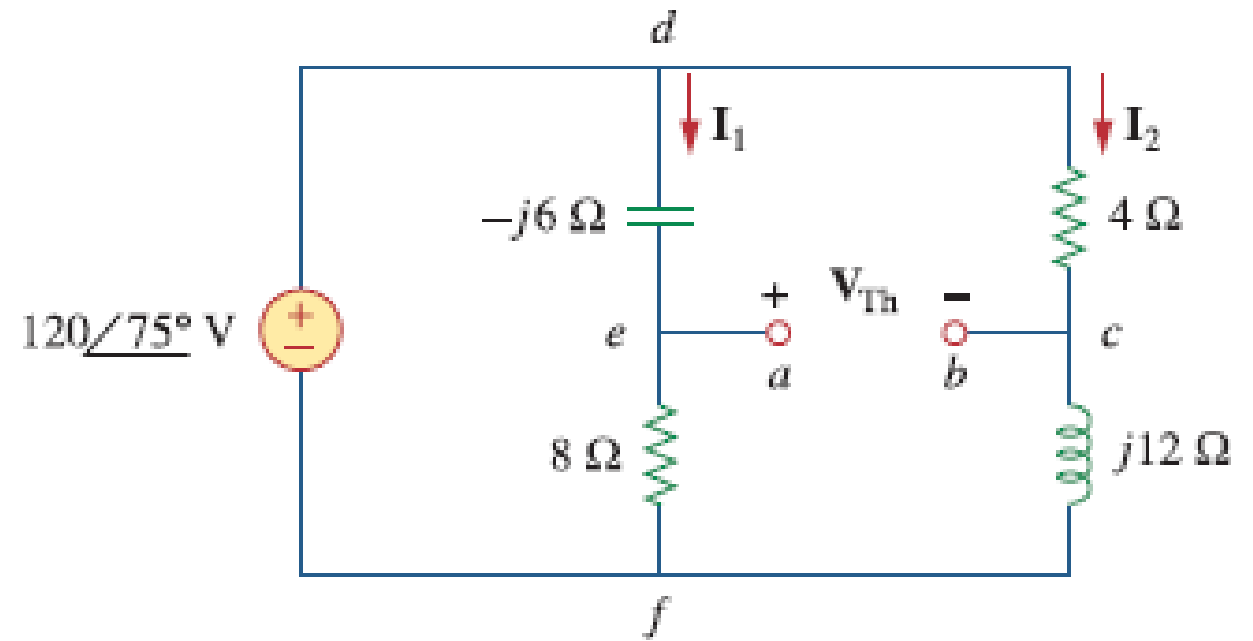
$$\mathbf{Z}_{Th} = \mathbf{Z}_1 + \mathbf{Z}_2 = 6.48 - j2.64 \, \Omega$$



- To find V_{Th} :

$$I_1 = \frac{120 \angle 75^\circ}{8 - j6} \text{ A,}$$

$$I_2 = \frac{120 \angle 75^\circ}{4 + j12} \text{ A}$$



Applying KVL around loop *bcdeab*

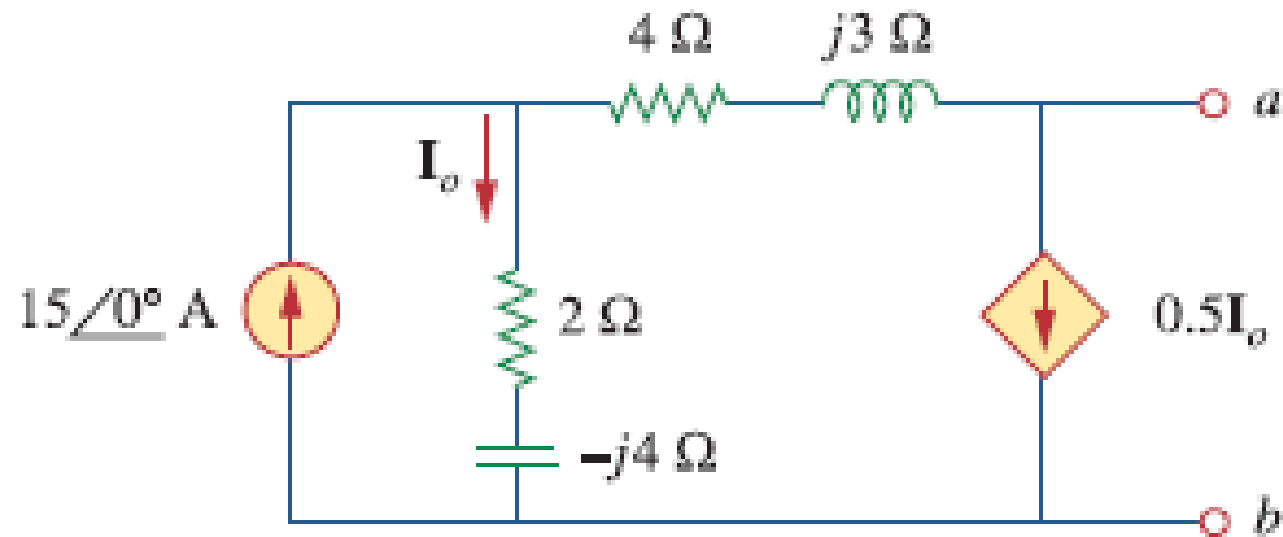
$$\begin{aligned} V_{Th} &= 4I_2 + j6I_1 = \frac{480 \angle 75^\circ}{4 + j12} + \frac{720 \angle 75^\circ + 90^\circ}{8 - j6} \\ &= 37.95 \angle 3.43^\circ + 72 \angle 201.87^\circ \\ &= -28.936 - j24.55 = 37.95 \angle 220.31^\circ \text{ V} \end{aligned}$$

Note: Another method:

Use voltage division at *a* and *b*, then subtract the results to get V_{th} .

Numerical -2

Find the Thevenin equivalent of the circuit



Solution:

- To find V_{Th} :

we apply KCL at node 1

$$15 = I_o + 0.5I_o$$

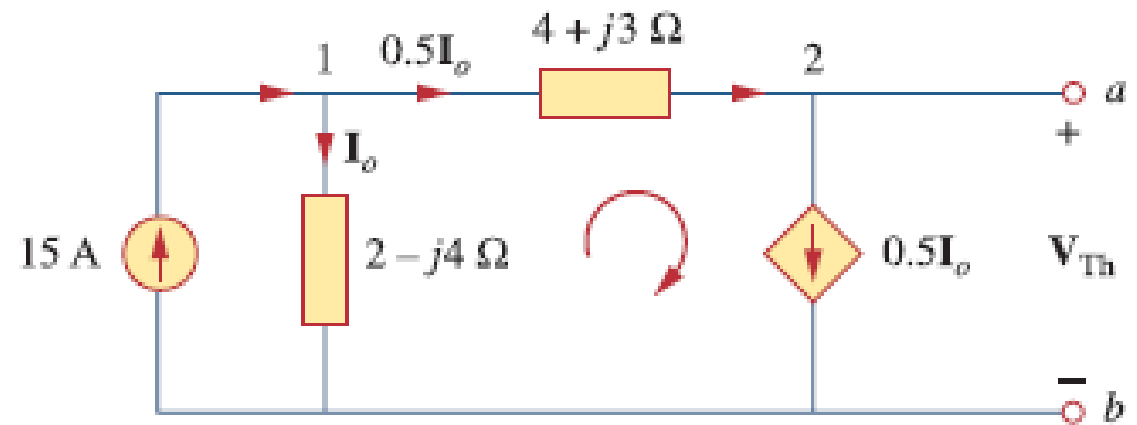
$$I_o = 10 \text{ A}$$

Applying KVL to the loop on the right-hand side

$$-I_o(2 - j4) + 0.5I_o(4 + j3) + V_{Th} = 0$$

$$V_{Th} = 10(2 - j4) - 5(4 + j3) = -j55$$

$$V_{Th} = 55 \angle -90^\circ \text{ V}$$



To find R_{Th} :

- Remove the independent current source
- Connect any arbitrary current source across the open end, find the voltage across it.
- Applying KCL at node V_s

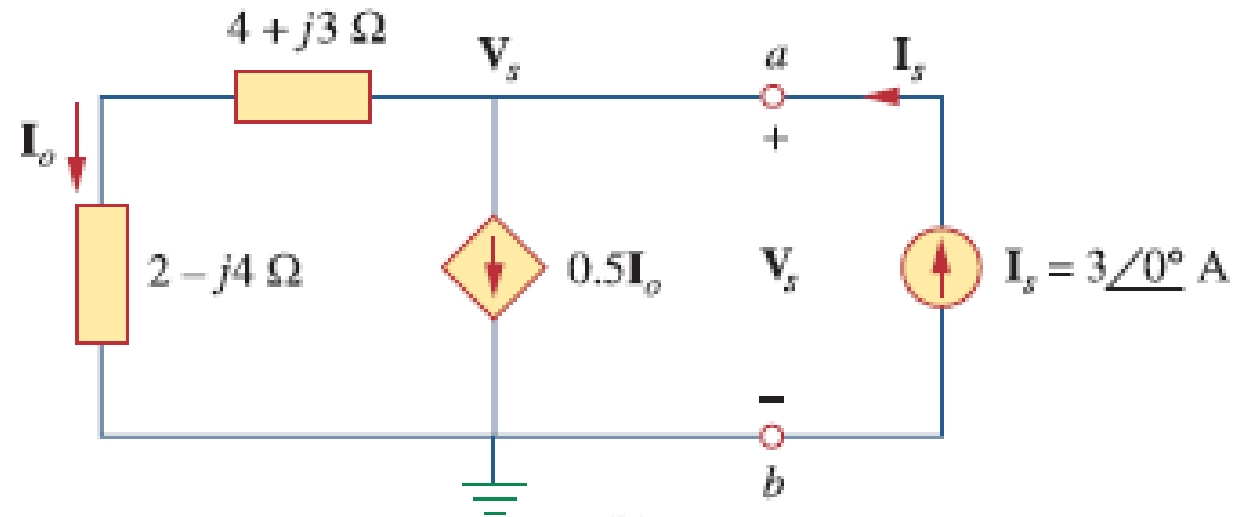
$$3 = I_o + 0.5I_o \Rightarrow I_o = 2 \text{ A}$$

Applying KVL to the outer loop

$$V_s = I_o(4 + j3 + 2 - j4) = 2(6 - j)$$

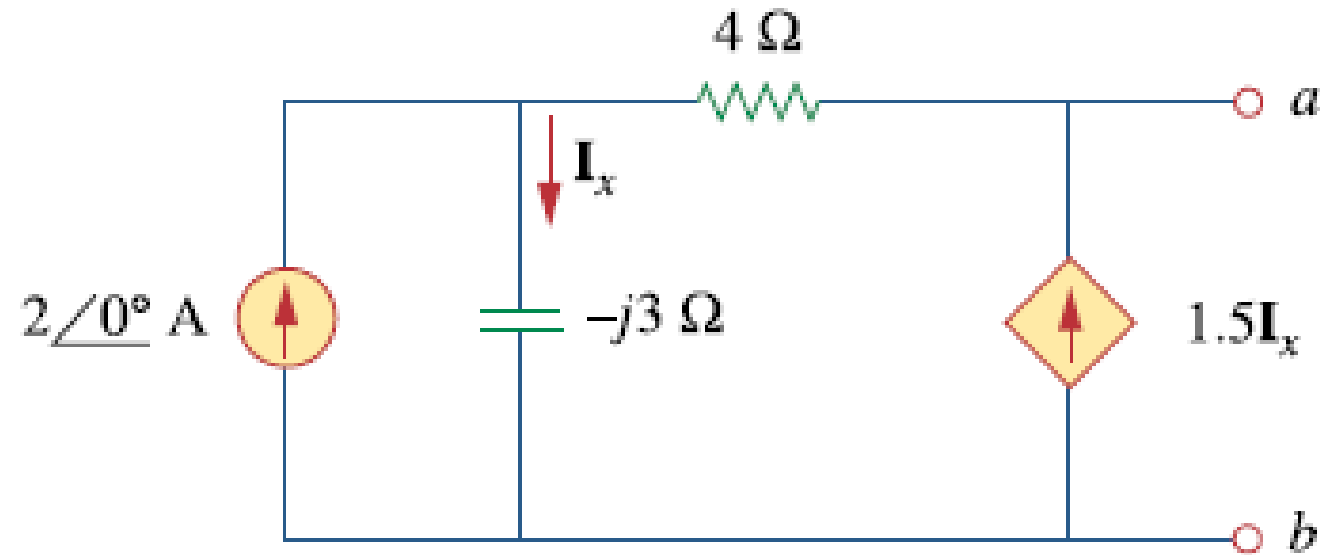
The Thevenin impedance is

$$Z_{Th} = \frac{V_s}{I_s} = \frac{2(6 - j)}{3} = 4 - j0.6667 \Omega$$

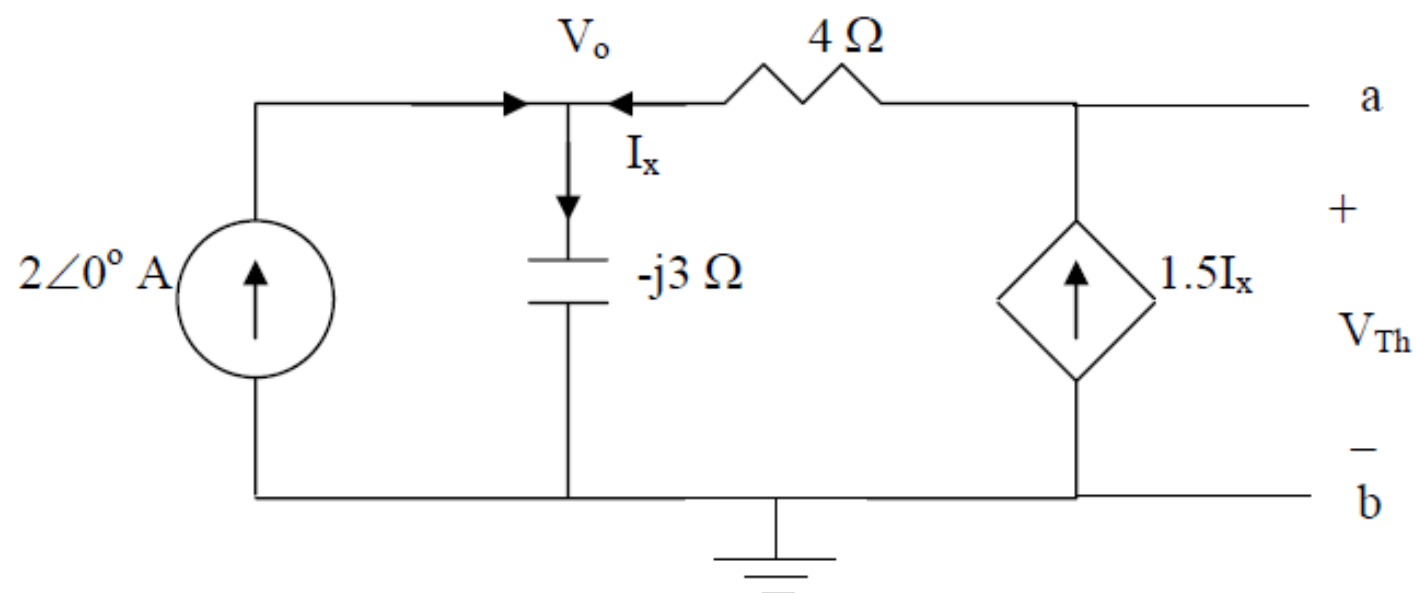


Numerical -3

Find the Thevenin equivalent at terminals a - b of the circuit



To find $\bar{V}_{Th}$, consider the circuit below

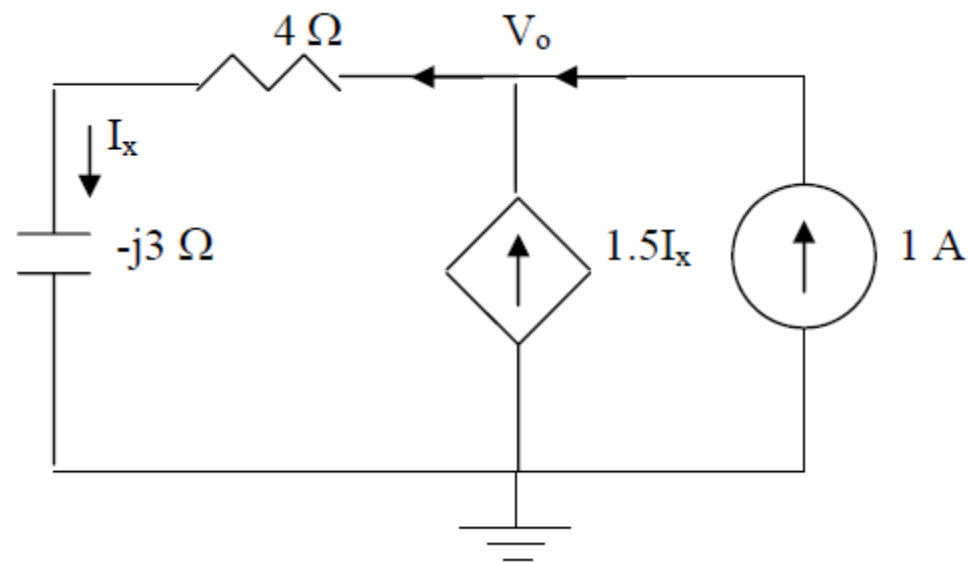


$$2 + 1.5I_x = I_x \longrightarrow I_x = -4$$

$$\text{But } V_o = -j3I_x = j12$$

$$V_{Th} = V_o + 6I_x = \underline{j12 - 24\text{ V}}$$

To find Z_{Th} , consider the circuit shown below.



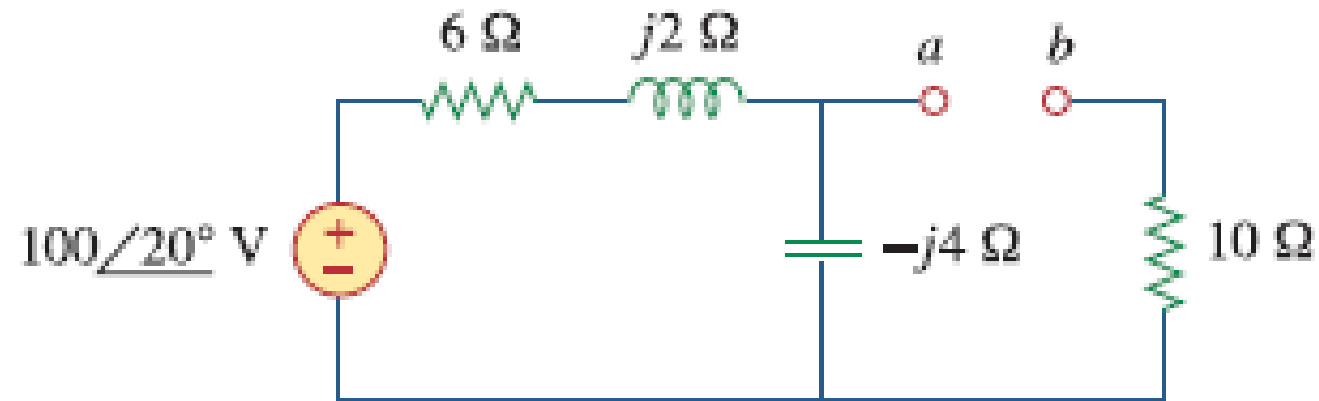
$$1 + 1.5 I_x = I_x \quad \Rightarrow \quad I_x = -2$$

$$-V_o + I_x(4 - j3) = 0 \quad \longrightarrow \quad V_o = -8 + j6$$

$$Z_{Th} = \frac{V_o}{1} = \underline{-8 + j6\ \Omega}$$

Homework problem -1

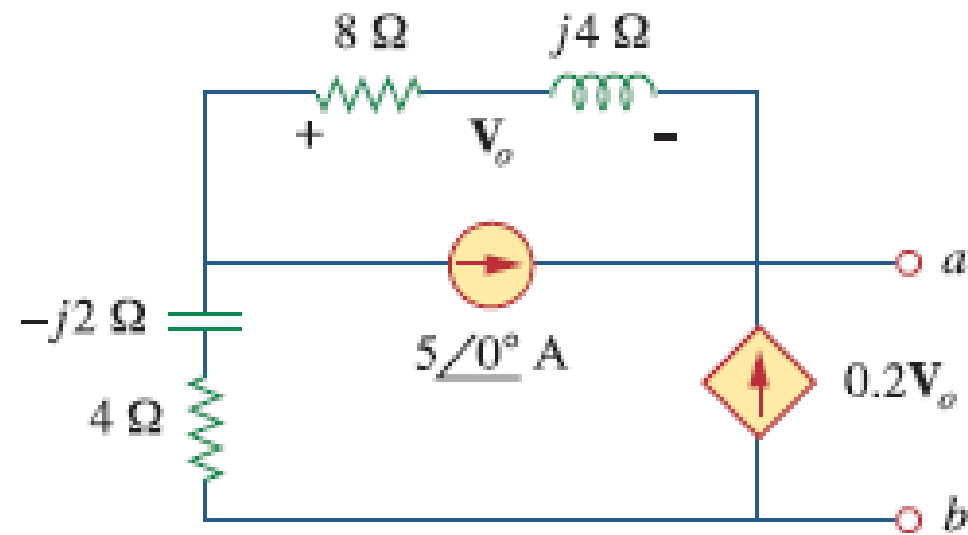
Find the Thevenin equivalent at terminals a - b of the circuit in Fig.



Answer: $Z_{Th} = 12.4 - j3.2\ \Omega$, $V_{Th} = 63.24\angle -51.57^\circ$ V.

Homework problem -2

Determine the Thevenin equivalent of the circuit in Fig.



Answer: $\mathbf{Z_{Th} = 4.473\angle -7.64^\circ\ \Omega}$, $\mathbf{V_{Th} = 7.35\angle 72.9^\circ\text{ V}}$.